

CS 221: Artificial Intelligence: Principles and Techniques

Homework 6: Bayesian Networks

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By turning in this assignment, I agree by the Stanford honor code and declare that all of this is my own work.

Before you get started, please read the Assignments section on the course website thoroughly.

Problem 1: Basic Bayes

Let's consider a simple scenario which we can model using a Bayesian network. You are walking to CS221 lecture and, on the way, you notice the grass on the Stanford Oval is wet. You reason that this could be because either it rained (rare in beautiful and sunny Palo Alto) or the sprinklers were on. Letting your mind wander from the lecture, you decide to sit down and figure out what happened here by applying your knowledge of Bayesian networks.

1a Bayesian Network Diagram

What we expect: A graph showing the causal relationships between variables. Use the variables Rain, Sprinkler, and WetGrass.

Your Response: Rain→WetGrass, Sprinkler→WetGrass

1b Joint Probability Expression

What we expect: The simplified joint probability expression, in terms of the local conditional probabilities (not a numerical value).

Your Response: $\mathbb{P}(\text{Rain}, \text{Sprinkler}, \text{WetGrass}) = \mathbb{P}(\text{Rain}) \cdot \mathbb{P}(\text{Sprinkler}) \cdot \mathbb{P}(\text{WetGrass} \mid \text{Rain}, \text{Sprinkler})$

1c Marginal Probability

What we expect: The simplified marginal probability expression for $\mathbb{P}(\text{WetGrass})$ in terms of local conditional probabilities.

Your Response: $\mathbb{P}(\text{WetGrass}) = \sum_{\text{Rain, Sprinkler}} \mathbb{P}(\text{Rain, Sprinkler, WetGrass})$

1d Likelihood Calculation

Assume $p(\text{Rain}) = 0.2$, $p(\text{Sprinkler}) = 0.5$, and the following local conditional probability table for $p(\text{WetGrass} \mid \text{Rain, Sprinkler})$:

Rain	Sprinkler	$p(\text{WetGrass} \mid \text{Rain, Sprinkler})$
True	True	0.9
True	False	0.8
False	True	0.7
False	False	0.0

What is $\mathbb{P}(\text{Rain} = \text{true}, \text{Sprinkler} = \text{true}, \text{WetGrass} = \text{false})$?

What we expect: A numerical answer with brief calculations shown.

Your Response:

$$\begin{aligned}
 \mathbb{P}(\text{Rain} = \text{true}, \text{Sprinkler} = \text{true}, \text{WetGrass} = \text{false}) &= \mathbb{P}(\text{Rain} = \text{true}) \cdot \mathbb{P}(\text{Sprinkler} = \text{true}) \\
 &\quad \cdot \mathbb{P}(\text{WetGrass} = \text{false} \mid \text{Rain} = \text{true}, \text{Sprinkler} = \text{true}) \\
 &= 0.2 \cdot 0.5 \cdot (1 - 0.9) \\
 &= 0.01
 \end{aligned}$$

1e Independence Analysis

Mark the following claims as true (T) or false (F):

- (a) Rain and Sprinkler are independent.
- (b) Rain and Sprinkler are conditionally independent given WetGrass.
- (c) $\mathbb{P}(\text{Rain} \mid \text{WetGrass}) > \mathbb{P}(\text{Rain} \mid \text{WetGrass}, \text{Sprinkler})$.

What we expect: Mark each statement T/F. No justification required.

Your Response:

1. T

2. F

3. T

Problem 2: Sampling

You get distracted while examining that luscious green grass in the Stanford Oval, and you somehow wander into the Gilbert Biological Sciences Building. As you realise where you are, you overhear some biologists discussing the genome of a new species they've discovered, provisionally named *Kenius bayus*. You can help them by modelling a phylogenetic tree with Bayesian networks where nucleotides mutate independently with probability `mutation_rate`.

2d Forward Sampling Table

Run `test_forward_sampling` with `mutation_rate = 0.1` and `genome_length = 10`. Record the sampled genomes.

Species	Genome
Thomas bayus	T
Humblus studentus	T
Aryamus bayus	T
Kenius bayus	T

What we expect: A completed table containing the sampled genome for each species.

The joint distribution probability is 18.2250000000%

2g Sampling Comparison

Using `test_gibbs_vs_rejection`, estimate $\mathbb{P}(\text{ThomasBayus} = \text{AAAC} \mid \text{KeniusBayus} = \text{AAAA})$.

Method	Steps / Iterations	Estimated Posterior
Rejection sampling	100	0.0000
Rejection sampling	10,000	0.0667
Gibbs sampling	100	0.0000
Gibbs sampling	10,000	0.0043
Exact	—	0.0335

What we expect: Populate the table with the posterior estimates for each method.

Problem 3: Learning

You are managing a team of annotators who label conversations as **good** or **bad**. For now, assume you have ground-truth labels.

3a Annotator Network Diagram

What we expect: Draw the Bayesian network relating Sample, Annotator1, Annotator2, and Annotator3.

Your Response: $Y \rightarrow A_1, Y \rightarrow A_2, Y \rightarrow A_3$

3b Perfect Annotator CPT

What we expect: Complete the CPT for a perfect annotator.

Sample	$p(\text{Annotator1} = \text{good} \mid \text{Sample})$	$p(\text{Annotator1} = \text{bad} \mid \text{Sample})$
good	1.0	0.0
bad	0.0	1.0

3c Random Annotator CPT

What we expect: Complete the CPT for an annotator who guesses uniformly at random.

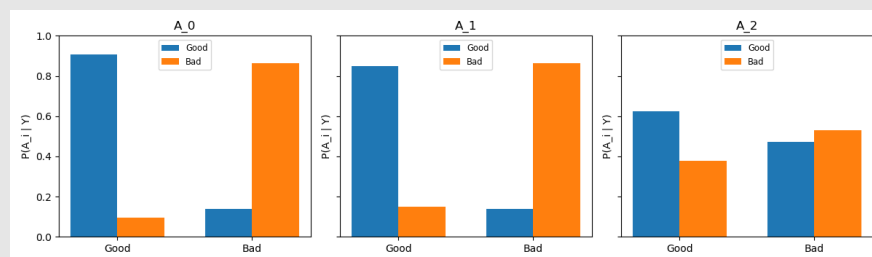
Sample	$p(\text{Annotator1} = \text{good} \mid \text{Sample})$	$p(\text{Annotator1} = \text{bad} \mid \text{Sample})$
good	0.5	0.5
bad	0.5	0.5

3f Annotator Analysis

After training with MLE, plot the annotator CPTs using `plot_annotator_cpts`. Which annotator appears least trustworthy, and why?

What we expect: Identify the least reliable annotator and justify using the plots.

Your Response:



It's easy to find that A_2 is the least trustworthy.

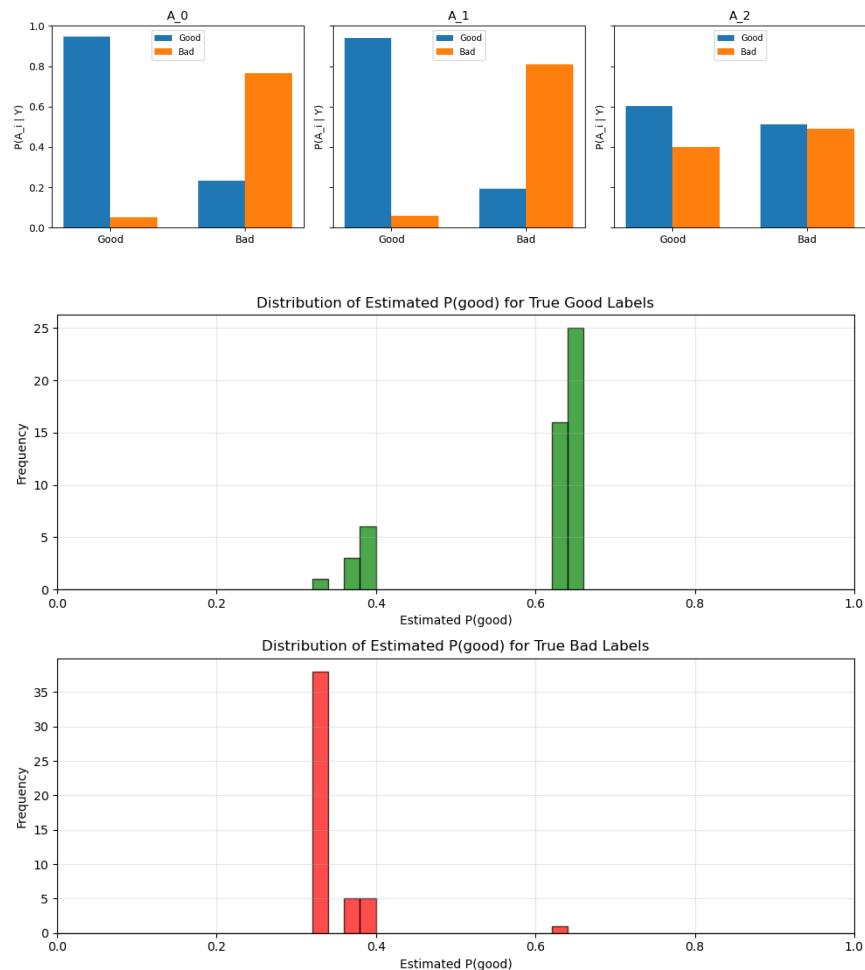
Problem 4: Learning without Labels

Now assume you do *not* have ground-truth labels. You will model both annotator noise and latent labels using the Expectation-Maximization algorithm.

4d EM Plots

Run `test_em_learn` for 100 iterations. Plot the annotator CPTs and the label prior using `plot_annotator_cpts` and `plot_label_cpt`. Include both plots below.

What we expect: Insert the two plots generated after EM training.



4e EM Recovery Analysis

Based on the plots, did EM recover the true labels? Explain your reasoning.

What we expect: State yes/no and support your answer using evidence from the plots.

Your Response: Yes, EM successfully recovered the latent labels. Although the ground truth labels were hidden, the final distribution of estimated $P(\text{good})$ shows a clear separation between the samples that were actually "good" (clustering around 0.6-0.7) and those that were actually "bad" (clustering around 0.3-0.4). The fact that the annotator CPTs converged to values reflecting different levels of accuracy (similar to the MLE results with labels) further indicates that the model has learned which annotators are more reliable and adjusted the latent label estimates accordingly.